Prospective SUM-159 WGD pilot observables

Specified on 23 October 2026, before SUM-159 WGD+/− motility data; no experimental observations were used.

Model drift subset: f ≤ 0.01 and multiple-domain fraction ≤ 0.05. In cells, classify by the observed frequency of coexisting protrusions; model f and native times have no calibrated cellular equivalents.

Cellular observable	Model counterpart	Drift regime: WGD+ / WGD−	Secondary-domain regime: WGD+ / WGD−	Mechanistic interpretation
Common-mode motion of the leading region's two lateral edges	Correlated boundary translation and domain drift (Figs 3D, 4D)	Similar (Holmes/Spring); smaller in Wave-pinning/Goryachev	No common direction identified outside persistent single-domain bouts	Translation; does not identify a Figure 2 route
Secondary-protrusion birth rate per unit perimeter	Persistent no-overlap birth rate (Fig. 4E(i))	Rare in both groups	Model dependent: Spring higher and Goryachev lower at f = 0.1; other trends unresolved	Initiation; model-dependent size response
Takeover incidence within a fixed follow-up window	Five-frame dominance within 20τ after birth (Fig. 4E(ii))	Not applicable when births are too rare	Lower by 20τ; persists through 80τ in the same cohort; lifetime fate unidentified	Incumbent resistance; lifetime success unresolved
Activity/protrusion-heading autocorrelation time	Heading decay and size exponent (Fig. 4B–D)	Longer; approximately size squared only when physical drift is size independent	Size advantage shrinks in the tested models; compare coherence and domain counts	Regime readout; no unique route assignment
Edge displacement after boundary-straddling photoactivation	Weak deterministic edge response (Fig. 3C)	Smaller (Wave-pinning/Goryachev); similar (Holmes/Spring)	Not specified: the pulse assay uses a stable polarized incumbent	Susceptibility within a route, not a unique mechanism

Prospective directions, not SUM-159 observations or a full statistical preregistration. Freeze sampling, segmentation, physical pulse dose and group inference before comparison; model f and native times have no calibrated cellular equivalents.

Birth rates are model dependent: at f = 0.1, bounds identify a Spring increase and a Goryachev decrease with size. Takeover within 20τ is fixed-horizon incidence; its decline persists through 80τ in the same cohort, while lifetime fate and certification selection remain unresolved.

Common translation tests the drift mechanism. Edge pulses refine susceptibility within routes: Wave-pinning and Holmes share an expansion-constrained class in Figure 2 but predict different edge responses. Combine these readouts with Figure 2 first-rise, width and pulse-gap assays to test initiation-limited reset, saturation/expansion and retained progress after visible activity loss. Activity heading, observed migration and OU-fitted memory remain separate readouts.